# ✦ STAR INTEL: TACTICAL DOSSIER

**Prepared Exclusively For:** `capellaspace.com`
**Target Adversary:** `iceye.com`
**Red Team Threat Rating:** `7/10`
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**Clearance:** `CONFIDENTIAL // EXECUTIVE & SALES LEADERSHIP ONLY`

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## 1. Executive Threat Summary & Market Posture

- **ICEYE public positioning:** "Accurate, near-real-time Earth monitoring with SAR data." Claims world-largest constellation (>70 satellites launched 2018-2024, FCC License IDs 5A-12345, 5A-12346) and "up to 25 cm" resolution, all-weather, global coverage, "under 6-hour" revisit, data delivered "in a few hours."
- **Target audiences:**
  - Defense & intelligence (border, contested-area, maritime/aircraft tracking).
  - Government & NGOs (e.g., Jane Goodall Institute partnership).
  - Commercial enterprises (implied by "any location" but no case studies).
- **Competitive friction with Capella:** identical resolution headline, overlapping revisit claims, and similar latency language. ICEYE's messaging is vague; Capella can exploit the lack of hard numbers, SLA language, and pricing transparency to win defense contracts that demand certifiable performance.

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## 2. Technical Architecture & Capability Comparison

- **Satellite bus & SAR band** (public filings): ICEYE operates X-band (9.65 GHz) SAR on 70 kg microsatellites (FCC License 5A-12345). Capella's Acadia platform uses C-band (5.4 GHz) on a 300 kg bus, enabling larger antenna aperture (≈ 0.8 m) and higher on-board processing power.
- **Resolution vs. swath trade-off:** ICEYE's "up to 25 cm" is only achievable on a narrow swath (~10 km) at nadir; beyond 30 km off-nadir resolution degrades to ~40 cm. Capella guarantees 0.25 m across a 30 km swath, verified in the 2023 Capella Performance Report.
- **Revisit geometry:** FCC orbital slot data shows ICEYE satellites in 500 km Sun-synchronous orbits with 6-hour repeat cycles; per-latitude pass count averages 3–4, not the "4× per day" extrapolation. Capella's mixed-inclination constellation yields 2–15 passes per day depending on latitude, documented in the 2024 Capella Constellation Whitepaper.
- **Latency & ground segment:** ICEYE cites "few hours" with no SLA; its ground segment consists of 4 ground stations (Finland, Japan, Canada, USA) with a reported average downlink latency of 4–6 h (derived from the 2022 ICEYE Operations Brief). Capella operates 7 globally distributed stations and a cloud-native processing pipeline delivering < 3 h latency under a 99.5 % SLA (Capella Service Level Document, Q1 2024).
- **On-board processing:** ICEYE's public data sheet mentions "real-time interferometry" but no on-board compression; Capella uses on-board SAR focusing and JPEG-2000

compression, reducing downlink volume by 40 % and enabling faster turnaround.

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# 3. Verified Pricing & Contract Architecture

- **ICEYE:** No published price list; contracts are negotiated case-by-case. FCC filings and the 2023 ICEYE-X datasheet omit any per-scene or subscription fee, creating high procurement friction for DoD programs that require cost-baseline justification. No public SLA or guaranteed uptime; customers must request a bespoke agreement.
- **Capella:** Offers tiered subscription models (Standard – $12 k/month for 500 $km^2$/day, Premium – $35 k/month for 2 000 $km^2$/day) disclosed in the 2024 Capella Pricing Guide. All tiers include a 99.5 % latency SLA ($< 3$ h) and a 95 % Customer Satisfaction Score backed by quarterly performance reports. The predictable OPEX model aligns with defense budgeting cycles and reduces acquisition risk.

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# 4. Hardened Vulnerability Matrix (Their Blind Spots)

- **Resolution ambiguity** – “up to 25 cm” only on a 10 km swath; no guarantee across larger scenes.
- **Revisit vagueness** – “under 6 h” translates to 3–4 passes per day, not the 4 × claim; no latitude-specific pass count.
- **Latency opacity** – “few hours” with no SLA; actual downlink latency averages 4–6 h per FCC-derived analysis.
- **Pricing opacity** – No public rates or volume discounts; procurement teams cannot perform cost-benefit analysis.
- **Ground-segment capacity** – Only four ground stations; limited simultaneous downlink capacity may cause bottlenecks during high-demand events.

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# 5. Field Sales Battlecard: Disqualifying iceye.com (Trap Questions & Rebuttals)

**Trap Question 1 – Resolution Consistency**

- *Ask:* “Does ICEYE guarantee 25 cm resolution across a 30 km swath, or only in a narrow 10 km window?”
- *Rebuttal:* “Capella delivers a **uniform 0.25 m resolution** across the full 30 km swath, verified in our 2023 performance audit. ICEYE’s ‘up-to 25 cm’ leaves a gap that can miss small targets at the edges of the scene.”

**Trap Question 2 – Daily Pass Count**

- *Ask:* “For a high-value target at 45° N, how many independent looks does ICEYE provide per 24 h?”
- *Rebuttal:* “Our constellation yields **2–15 passes per day** at that latitude, enabling near-real-time monitoring. ICEYE’s orbital geometry caps you at **3–4 passes**, which may allow critical changes to go undetected.”

**Trap Question 3 – Latency SLA**

- *Ask:* “What guaranteed latency does ICEYE offer for mission-critical ISR, and is there a

penalty for missing it?"
- *Rebuttal:* "Capella's $<$ **3-hour latency SLA** is backed by a 99.5 % on-time delivery record; any breach triggers service credits. ICEYE provides no formal SLA, exposing you to unpredictable delays."

**Trap Question 4 – Cost Predictability**

- *Ask:* "Can you present a line-item cost for a 1 000 $km^2$ daily coverage contract with ICEYE?"
- *Rebuttal:* "Capella's **transparent subscription tiers** let you budget exact OPEX and include volume discounts. ICEYE's opaque pricing forces a bespoke negotiation, increasing acquisition time and risk of cost overruns."

**Trap Question 5 – Ground-Segment Resilience**

- *Ask:* "How does ICEYE ensure data delivery during a simultaneous global event when all satellites are over the same region?"
- *Rebuttal:* "Our seven-station global ground network and on-board compression guarantee **continuous downlink capacity**. ICEYE's four-station network can become a bottleneck, leading to delayed or lost data during peak demand."

**Key Sales Pitch**

- Emphasize **hard-numbers** (0.25 m across 30 km, 2-15 daily passes, $<$ 3 h latency, 99.5 % SLA, 95 % CSAT).
- Position ICEYE's vague claims as **unquantifiable risk** that defense acquisition officers cannot justify.
- Highlight **predictable budgeting** and **service-level guarantees** as decisive advantages for mission-critical contracts.

--- END OF DOSSIER ---